COMPACTLY SUPPORTED COHOMOLOGY SENDS DISTINGUISHED TRIANGLES TO INVERSE SYSTEMS OF TRIANGLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G52 (sources/stacks-duality/pinned-original-source.tex line 8761).

Allowed standard prerequisites (editorial).

- Inverse systems of triangles for an open immersion. Same chapter, Lemma labelled lemma-extension-by-zero-triangle, cited at proof lines 8780-8784. Supplies the system of triangles on the compactification.
- The definition of compactly supported cohomology. Same chapter, Situation labelled situation-shriek, cited at proof lines 8760-8774 and 8788-8789. Identifies the pushed forward pro-systems with Rf_! of the three complexes.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G4M: sources/stacks-duality/pinned-original-source.tex, source lines 8145–8145. Same-chapter pointer into the author environment lemma at source line 8145; invoked at proof line 8784 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. Let*

$$K \to L \to M \to K[1]$$

be a distinguished triangle of $D^b_{Coh}(\mathcal{O}_X)$. Then there exists an inverse system of distinguished triangles

$$K_n \to L_n \to M_n \to K_n[1]$$

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in $D^b_{Coh}(\mathcal{O}_Y)$ such that the pro-systems (K_n), (L_n), and (M_n) give $Rf_!K$, $Rf_!L$, and $Rf_!M$.

Proof. Choose a compactification $j : X \to \overline{X}$ over Y and denote $\overline{f} : \overline{X} \to Y$ the structure morphism. Choose an inverse system of distinguished triangles

$$\overline{K}_n \to \overline{L}_n \to \overline{M}_n \to \overline{K}_n[1]$$

in $D^b_{Coh}(\mathcal{O}_{\overline{X}})$ as in Lemma 0G4M corresponding to the open immersion j and the given distinguished triangle. Take $K_n = R\overline{f}_*\overline{K}_n$ and similarly for L_n and M_n. This works by the very definition of $Rf_!$. □